MAY 4 – 8, 2026

Science Team Weekly

New Bedford High School · Science Department

PROFESSIONAL LEARNING COMMUNITY

Three Sessions, One Arc: Mapping, Internalizing & Co-Planning

Science Department PLC Recap — May 5, 6, and 8, 2026

This week's PLC brought the Science Department through a carefully sequenced three-session arc designed to move teachers from **data to planning to practice**. Rather than treating each session as a standalone meeting, the arc was built as a coherent professional learning sequence: each session built directly on the previous, culminating in a co-planned lesson ready for immediate implementation.

TUESDAY, MAY 5 · 46 MINUTES

Session 1 — Lesson Mapping: Standards, Pacing & Purpose

Teachers opened by reviewing Q3 benchmark data silently, identifying which standards showed the greatest need for instructional attention. Each teacher then completed a **lesson map** covering the next 2–3 weeks — every lesson required a standard, a clear purpose, and a mastery indicator. The session closed with a team share-out and a written commitment to one targeted instructional adjustment.

Q3 Data Review · Lesson Mapping · Standards Alignment · Team Share-Out · Written Commitment

WEDNESDAY, MAY 6 · 46 MINUTES

Session 2 — Deep Lesson Internalization

Teachers selected one upcoming lesson and completed the **key task or exit ticket exactly as a student would** — without reading ahead or planning. They then annotated the teacher edition with facilitation notes, focusing on the critical moment of student thinking. The final segment addressed access and equity planning, identifying potential barriers and scaffolds before closing with preparation for Friday's co-planning session.

Do the Student Work · Lesson Annotation · Access & Equity · Prep for Co-Planning

FRIDAY, MAY 8 · 41 MINUTES

Session 3 — Co-Planning: Building the Lesson Together

The classroom teacher walked the co-teacher through the lesson map and an annotated overview of the upcoming lesson. Together, they identified the **three most critical moments** where the co-teacher's presence would have the greatest impact — small-group facilitation, targeted questioning, or differentiated support. They assigned specific roles, planned differentiation strategies for identified students, and closed with written commitments for implementation and a brief reflection protocol.

Lesson Sync · Co-Teacher Roles · Differentiation Planning · Written Commitments

THE POWER OF THE THREE-SESSION ARC

Lesson mapping anchors instruction to standards and student data — not just pacing guides. Internalization builds teacher empathy for the student experience *before* facilitation begins. Co-planning ensures both educators have active, meaningful roles in every lesson. Each arc ends with a written commitment, turning reflection into action.

THIS WEEK IN CLASSROOMS

CLASSROOM HIGHLIGHTS

- **Biology Enrichment:** Cassidy Barrett's "Battle of the Beaks" natural selection simulation — students model adaptation and competition
- **Pre-AP Chemistry:** Samantha Roche's molar mass of compounds and The Mole POGIL — collaborative inquiry with periodic tables
- **Biology Lab:** Brian McCauley's osmosis and cell transport lab — red onion cells under the microscope in two solutions
- **Science Corner:** Epigenetics — real-time dynamics of DNA methylation and histone acetylation

GUIDING QUESTION FOR THE ARC

How do we use student data and lesson internalization to plan instruction that reaches every learner? This question anchored all three sessions and will continue to guide our work as we move into the final stretch of the school year.

NEXT STEPS

Bring student work samples to the next PLC session. Implement your May 8 co-planned lesson and reflect on co-teacher role effectiveness. Consider: what did you notice about student thinking that you couldn't have anticipated without internalizing the lesson first?

TEACHER SPOTLIGHT · BIOLOGY ENRICHMENT

Battle of the Beaks: Natural Selection in Action

Ms. Cassidy Barrett's Biology Enrichment — Natural Selection Simulation

MS. CASSIDY BARRETT — BIOLOGY ENRICHMENT

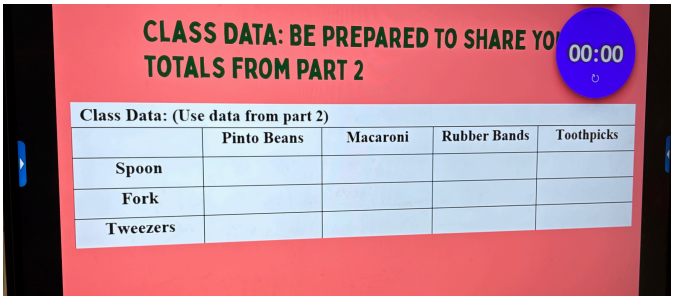
This week, students in Biology Enrichment explored **natural selection, adaptation, and competition** through a hands-on simulation called *Battle of the Beaks*. Each student was assigned a "beak type" — spoon, fork, or tweezers — and competed to collect food items (pinto beans, macaroni, rubber bands, and toothpicks) from a shared feeding area, recording how much food each beak type could gather in a timed round. The simulation ran across two rounds, with the second round introducing multiple food types simultaneously to model a more complex and competitive habitat.



Students use tweezers, clothespins, and spoons to collect food items from a shared plate, simulating how beak morphology determines feeding success in a competitive habitat.

Each student held their "beak" in one hand and a cup (the "stomach") in the other — only one food item could be moved at a time. When time ran out, students counted and recorded their totals in a class data table, then pooled results to analyze which beak type was most successful with each food type. **Part 2 introduced direct competition**: all beak types competed simultaneously for all food types, forcing students to develop a strategy and observe how resource availability drives differential survival.

After the simulation, students answered reflection questions connecting their data to core evolutionary concepts: Which beak collected the most food? Which would survive a drought that eliminated rubber bands? What would happen if only toothpicks remained? These questions pushed students to **apply natural selection logic** to novel scenarios — exactly the kind of higher-order thinking that bridges lab experience to conceptual understanding of Darwin's theory.



The class data table projected on screen: each beak type's totals for Pinto Beans, Macaroni, Rubber Bands, and Toothpicks are compiled and shared for group analysis.

SIMULATION MATERIALS

Beak types: Spoons, forks, and tweezers represented different beak morphologies. **Food types:** Pinto beans, macaroni, rubber bands, and toothpicks modeled different food sources. Students used cups as "stomachs" and a stopwatch to enforce timed rounds of competition.

WHAT ARE "MONEY QUESTIONS"?

After the simulation, Ms. Barrett pushed students beyond data collection with high-leverage questions: *"Which beak would be eliminated if rubber bands disappeared?"* and *"What would happen to the population if only one food type remained?"* These questions require students to apply, analyze, and explain — connecting simulation results directly to the mechanisms of natural selection and differential reproduction.

KEY EVOLUTIONARY CONCEPTS

Natural Selection: Individuals with traits better suited to the environment survive and reproduce more successfully. **Adaptation:** Beak morphology represents an inherited trait shaped by selective pressure over generations. **Competition:** Multiple species competing for the same limited resource is a core driver of evolutionary change. **Fitness:** The beak that collects the most food has the highest fitness and survives to pass on its traits.

PLC CONNECTION

This lesson reflects the department's PLC focus on lesson internalization. **Ms. Barrett experienced the student task herself** before facilitating — understanding the physical challenge of each beak type informed her facilitation and helped her anticipate where students would need support.

TEACHER SPOTLIGHT · PRE-AP CHEMISTRY

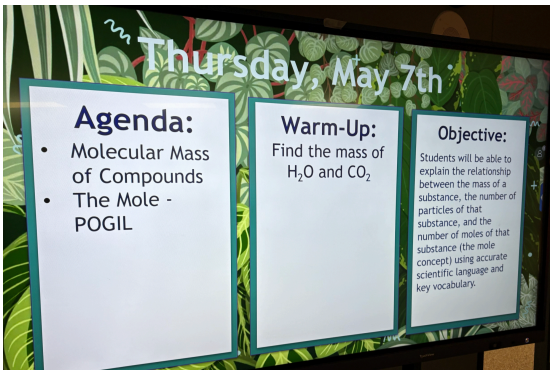
Molar Mass, The Mole, and the Power of POGIL

Ms. Samantha Roche's Pre-AP Chemistry — Molar Mass of Compounds & The Mole POGIL

MS. SAMANTHA ROCHE — PRE-AP CHEMISTRY

On Thursday, May 7th, students in Pre-AP Chemistry engaged in a two-part lesson designed to build both procedural fluency and conceptual understanding of one of chemistry's most fundamental ideas: **the mole**. The lesson opened with a warm-up that asked students to find the molecular mass of water (H_2O) and carbon dioxide (CO_2) using the periodic table — a task that required students to interpret chemical formulas, identify subscripts and coefficients, and apply atomic mass values with precision.

The warm-up was carefully scaffolded. Students worked through a structured worksheet that walked them through the rules for interpreting chemical formulas: how subscripts indicate the number of atoms of each element, how parentheses and coefficients multiply subscripts, and how to use the periodic table to find atomic mass values. By the time students reached the practice problems, they had a clear framework for calculating molar mass systematically — a skill that underpins stoichiometry, solution chemistry, and virtually every quantitative topic in chemistry.



Ms. Roche's Thursday, May 7th agenda slide: Molecular Mass of Compounds warm-up (H_2O and CO_2), followed by The Mole POGIL collaborative inquiry activity.



Students work collaboratively with periodic tables and reference sheets during the POGIL activity, pointing to specific elements and discussing atomic mass values.

WHAT IS POGIL?

Process Oriented Guided Inquiry Learning (POGIL) is a collaborative, inquiry-based instructional strategy in which students work in structured groups to construct understanding through guided questions. Rather than receiving information passively, students explore a model, answer guided questions, and build toward conceptual understanding through discussion and reasoning. POGIL is widely used in AP and Pre-AP Chemistry because it develops both content knowledge and scientific process skills simultaneously.

THE MOLE: WHY IT MATTERS

The mole is chemistry's counting unit — one mole of any substance contains 6.022×10^{23} particles (Avogadro's number). The lesson's objective asked students to explain the relationship between the mass of a substance, the number of particles, and the number of moles — using accurate scientific language and key vocabulary. This conceptual bridge is essential for all quantitative chemistry that follows.

LESSON OBJECTIVE

Students will be able to **explain the relationship between the mass of a substance, the number of particles of that substance, and the number of moles of that substance** (the mole concept) using accurate scientific language and key vocabulary. The warm-up on molar mass of H_2O and CO_2 directly supported this objective by grounding the abstract mole concept in concrete calculation.

PLC CONNECTION

This lesson reflects the department's PLC work on **lesson internalization and co-planning**. Ms. Roche's structured warm-up and POGIL sequence demonstrate how a well-internalized lesson creates natural entry points for co-teacher support — the collaborative group structure of POGIL allows a co-teacher to circulate, facilitate questioning, and provide targeted support to students who need scaffolding.

TEACHER SPOTLIGHT · BIOLOGY LAB

Stations, Questions, and the Power of Inquiry

Mr. Brian McCauley's Biology — Osmosis & Cell Transport Microscope Lab

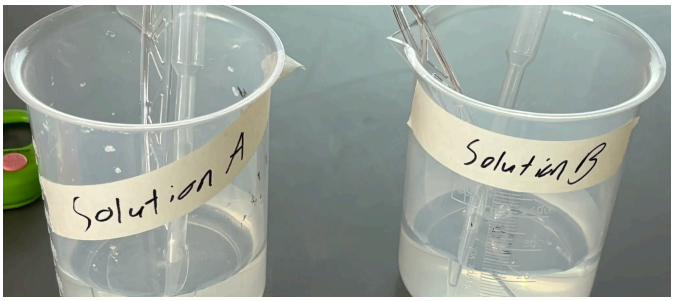
MR. BRIAN McCAULEY — BIOLOGY

This week, Mr. McCauley's Biology students investigated **osmosis and cell transport** through a microscope lab that placed the scientific process of observation at the center of the learning experience. Students examined red onion cells mounted in two different solutions — Solution A (hypotonic) and Solution B (hypertonic) — and used a structured **"Notice and Wonder"** framework to record their observations before drawing any conclusions. The lab was designed to let students encounter the phenomenon first and construct explanations from evidence, rather than receiving definitions before observation.

Each microscope station was labeled with its solution type, and students rotated through the stations in small groups, sketching what they saw and recording their observations in a two-column chart. The Notice and Wonder format — projected on the board at the start of class — asked students to separate what they could directly observe from what they were curious about or uncertain about. This distinction is a foundational scientific thinking skill: the ability to distinguish between observation and inference.



A student photographs her microscope slide through the eyepiece, capturing red onion cells in Solution B — documenting observations for the Notice and Wonder chart.



Solution A and Solution B, each with a dropper for slide preparation. Students placed onion cells in each solution and observed the effect of solute concentration on cell shape under the microscope.



A microscope station labeled "Solution B" alongside a student's Chromebook. Students used their devices to look up vocabulary and compare their observations to diagrams of plasmolysis and turgidity.

NOTICE AND WONDER FRAMEWORK

Students drew a two-column chart in their notebooks: **Notice** (direct observations about cell shape, color, and structure) and **Wonder** (questions and inferences about why the cells looked different in each solution). A third section, **Observations**, captured sketches of what students saw under the microscope. This structure separates empirical observation from interpretive reasoning — a key scientific practice.

WHAT THIS MEANS FOR YOUR CLASSROOM

Mr. McCauley's lab is a model of *phenomenon-first* instruction: students encounter the observable effect of osmosis before they receive the vocabulary or explanation. This approach mirrors the way scientists actually work — observation precedes theory. Consider asking students: *"What do you notice about the cell shape in each solution?"* before introducing the terms hypotonic, hypertonic, and isotonic.

PLC CONNECTION: QUESTIONING STRATEGIES

From PLC to Practice: The Notice and Wonder format is a direct application of the questioning strategies explored in our PLC sessions — designing questions that drive academic discourse and require evidence-based reasoning. **Osmosis + CER:** Students used the Claim-Evidence-Reasoning framework to structure their written responses, connecting observation, data, and scientific explanation in one seamless activity.

SCIENCE CORNER

The Epigenome Is Always Listening

Real-Time Dynamics of DNA Methylation, Histone Acetylation, and the Challenge of Watching the Genome Change Live

Every breath you take, every meal you eat, every moment of stress or joy — the epigenome is listening and responding. Unlike the DNA sequence itself, which remains essentially fixed from conception, the **epigenome** is a dynamic layer of chemical modifications that sits on top of the genome, controlling which genes are switched on or off, and when. Two of the most studied mechanisms are **DNA methylation** (the addition of methyl groups to cytosine bases at CpG dinucleotides, which typically silences gene expression) and **histone acetylation** (the addition of acetyl groups to histone tail proteins, which loosens chromatin structure and activates transcription). Together, these modifications form a molecular language that translates environmental experience into biological response.[1,2]

What makes epigenetics so scientifically compelling — and so challenging — is that the epigenome changes continuously. Histone acetylation can shift within minutes in response to a stimulus; DNA methylation patterns can be remodeled over hours or days by diet, stress, toxins, or exercise. Yet for decades, scientists faced a fundamental paradox: the very act of measuring the epigenome required destroying the cells being studied. Traditional methods like bisulfite sequencing, ChIP-seq, and ATAC-seq are **endpoint assays** — they require cells to be fixed or lysed, yielding only a static snapshot of a process that is inherently dynamic.[3,4]

SCIENTIFIC HIGHLIGHTS: LIVE-CELL EPIGENETIC SENSORS

Genetically encoded sensors (GECs): Fluorescent protein constructs that report epigenetic state in real time without disrupting the cell. **FRET-based biosensors:** Detect changes in histone modification status through fluorescence resonance energy transfer. **CRISPR/dCas-derived probes:** Use catalytically dead Cas9 fused to fluorescent reporters to track specific genomic loci. **Mintbodies:** Intracellular antibody fragments that bind modified histones and report their location in living cells. These tools are now being deployed in 3D organoid models to capture spatially distinct epigenetic states across tissue layers.[1]

A landmark 2026 study published in *Biosensors* by Tariq et al. describes a new generation of **live-cell epigenetic sensors** that are beginning to crack this problem. These tools — including genetically encoded sensors, FRET-based biosensors, and CRISPR/dCas-derived probes — allow scientists to track DNA methylation and histone modifications in real time in living cells, without fixation or lysis. Crucially, these sensors can now be deployed in three-dimensional organoid models, capturing not just temporal dynamics but also the *spatial* heterogeneity of epigenetic states across different regions of a tissue.[1]

The implications extend far beyond the laboratory. Epigenetic changes can be inherited across generations — a phenomenon known as **transgenerational epigenetics** — meaning that the environmental exposures of parents and grandparents may leave molecular marks on the genomes of their descendants.[5,6] This finding has profound implications for how we understand the relationship between nature and nurture: genetics provides the blueprint, but the environment writes the annotations. As Banushi et al. (2025) put it, epigenetics is the bridge between nature, nurture, and neurobiology — the mechanism by which lived experience becomes biological reality.[3]

CONNECTING TO YOUR CLASSROOM

Epigenetics is a natural anchor for multiple science courses. In **AP Biology**, it connects to gene regulation and the control of transcription. In **Environmental Science**, it illustrates how pollution, diet, and stress leave molecular marks on the genome. In **Forensics**, epigenetic clocks can estimate biological age from a DNA sample. Consider asking students: *"If the environment can change how genes are expressed, does that change how you think about the relationship between genetics and behavior?"*

SOURCES & FURTHER READING

[1] Tariq A, Naz I, Arshad F, et al. "Real-Time and Spatially Resolved Epigenetic Dynamics Tracking Beyond DNA Methylation via Live-Cell Epigenetic Sensors in 3D Systems." *Biosensors.* 2026;16(4):188.

[2] Allis CD, Jenuwein T. "The molecular hallmarks of epigenetic control." *Nature Reviews Genetics.* 2016;17:487–500.

[3] Banushi B, et al. "Epigenetic Echoes: Bridging Nature, Nurture, and Neurobiology." *PMC/NIH.* 2025.

[4] Benčik I, et al. "From DNA Methylation and Histone Modifications to Non-Coding RNAs: Evaluating Tools for Epigenetic Research." *Applied Sciences.* 2025;15(18):9940.

[5] Steg LC, et al. "Symposium summary: epigenetic inheritance — impact for biology and evolution." *Environmental Epigenetics.* 2024;10(1):dvae002.

[6] Faraji J, et al. "Environmental epigenetics: new horizons in redefining the relationship between the genome and environment." *Science of the Total Environment.* 2026.

[7] Iyer KA, et al. "The Future of Epigenetics: Emerging Technologies and Clinical Applications." *ACS Pharmacology & Translational Science.* 2026.

[8] Jaenisch R, Bird A. "Epigenetic regulation of gene expression: how the genome integrates intrinsic and environmental signals." *Nature Genetics.* 2003;33(Suppl):245–254.